

Syed Murtuza Quadri

syed.murtuza97@gmail.com
+49 1746708727
Pariser Straße, Kaiserslautern, Germany

Portfolio: syedmurtuzaquadri.com
linkedin.com/in/syed-murtuza-quadri



EDUCATION

M.Sc. Commercial Vehicle Technology
RPTU Kaiserslautern, Germany 10/2024 - Present
Grade: 2.3

B.E. Mechanical Engineering
Osmania University, India 2016 - 2020
Grade: 2.1

WORK EXPERIENCE

Door/Closure Systems Design Engineer
Maruti Suzuki India Ltd. | India 05/2022 - 08/2024
Research and Development (Body & Exterior)

- Engineered structural BIW (Body-In-White) side door components (door panels, beams, reinforcements) and integrated key mechanisms (hinges, latches, glass, regulators) in Siemens NX, for a passenger vehicle project.
- Validated door system performance through detailed sectional analysis for shutline/sealing feasibility and FEM analysis (NX Nastran) for structural integrity and closing dynamics.
- Collaborated cross-functionally with manufacturing (press, weld, paint) and styling teams to optimize designs, balancing manufacturability, cost, weight, safety, and DVP targets.

Mechanical Design Engineer
Emflux Motors Pvt. Ltd. | India 02/2021 - 04/2022

- Led the end-to-end delivery of two complex CNC automation systems: a stator winding machine and an EV battery spot-welding unit—managing the complete project lifecycle from concept and procurement through to final assembly and testing.
- Responsible for the comprehensive mechanical design in SolidWorks (1300+ components), applying advanced analysis including GD&T, tolerance stack-up, and FEA (ANSYS) to ensure precision and system robustness.
- Engineered the full electrical and controls system, from schematic design and sensor integration to CNC controller programming (G-code) and EMI mitigation.

ACADEMIC PROJECTS

Sensor Signal Processing (Grade 1.0)
Semester Project 04/2025 - 09/2025

- Real-Time voice command recognition system using sensor signal processing techniques in Python.
- Implemented FFT and Spectrogram feature extraction and multi-model classification (SVM) Support Vector Machine / (KNN) k-Nearest Neighbour/ Decision-Tree classifier for five speech commands.

Bachelor's Thesis
Composite Material Testing (GFRP) 2019 - 2020

- Title: A Study on the Changes in Flexural Properties of Eglass/ Epoxy Laminate Composite with Artificial Delamination.

EXTRA-CURRICULAR ACTIVITIES

Formula Student - Technical Team Member
Kaiserslautern Racing Team 09/2024 - 01/2025

- Part of the technical team, supporting the team till the FS Quiz round. Team qualified for FS Spain.

SAE BAJA - Chief Steering Engineer
Mudbrothers Racing (Off-Road) 02/2019 - 01/2020

- Designed the complete steering system with kinematic and dynamic analysis for an Off-Road vehicle.
- Led manufacturing of all the components in-house, with design optimization using Ansys Workbench.
- The team ranked 7th nationally in the endurance race.

Vice President
SAE India Collegiate Club 2018 - 2019

- Organized and led a two-day 'Engine Assembly Workshop,' single-handedly instructing 60 students in the complete hands-on assembly of a car engine.

SKILLS

- CAD:** Siemens NX, SolidWorks, PTC Creo, CATIA V5, AutoCAD
- CAE:** ANSYS Workbench, NX Nastran, MATLAB/Simulink, Signal Processing.
- Manufacturing:** CAM & CNC Machining, FDM/SLA 3D Printing, GD&T, DFA/DFM, Fault Tree Analysis, DFMEA
- Programming:** Python (NumPy, Pandas, sklearn), C++ (OOP), G-Code.

LANGUAGES

English	German	Hindi
C1 CEFR	A2 CEFR	Proficient

HOBBIES & INTERESTS

- Played for 'Hyderabad Sporting Football Club'.
- Proficient in portrait sketching and acrylic painting.
- Beginner guitarist, learning chords and strumming techniques.